PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. A body of mass 4 kg is placed on a plane at a point P having coordinates (3, 4) m. Under the action of force $\vec{F}=\left(2\hat{i}+3\hat{j}\right)$ N, it moves to a new point Q having coordinates (6, 10) m in 4 sec. The average power and instantaneous power at the end of 4 sec are in the ratio of

(1) 1 : 2 (2) 6 : 13

(3) 4 : 3 (4) 13 : 6

Answer (2)

Sol. $\vec{F}=\left(2\hat{i}+3\hat{j}\right)$ N ; displacement $\vec{S}=\left(3\hat{i}+6\hat{j}\right)$ m

Acceleration, $\vec{a}=\frac{\vec{F}}{m}=\left(\frac{2\hat{i}+3\hat{j}}{4}\right)$m/s^2

$$\vec{S}=\vec{u}t+\frac{1}{2}\vec{a}t^2$$

$$\Rightarrow \quad 3\hat{i}+6\hat{j}=4\vec{u}+\frac{1}{2}\left(\frac{2\hat{i}+3\hat{j}}{4}\right)\cdot 16=4\vec{u}+4\hat{i}+6\hat{j}$$

or $\vec{u}=\frac{1}{4}\left(3\hat{i}+6\hat{j}-4\hat{i}-6\hat{j}\right)$

$$\vec{u}=-\frac{1}{4}\hat{i}$$

Average Power $=\vec{F}\cdot\frac{\vec{S}}{t}$

$$=\frac{\left(2\hat{i}+3\hat{j}\right)\cdot\left(3\hat{i}+6\hat{j}\right)}{4}\text{ watt}$$

= 6 watt

Instantaneous Power $=\vec{F}\cdot\vec{v}$

$$=\vec{F}\cdot\left(\vec{u}+\vec{a}t\right)$$

$$=\left(2\hat{i}+3\hat{j}\right)\cdot\left(-\frac{1}{4}\hat{i}+2\hat{i}+3\hat{j}\right)$$

$$=\left(2\hat{i}+3\hat{j}\right)\cdot\left(+\frac{7}{4}\hat{i}+3\hat{j}\right)$$

= (3.5 + 9) watt

= 12.5 watt

27. A 400 g solid cube having an edge of length 10 cm floats in water. How much volume of the cube is outside the water?

(Given : density of water = 1000 kg m^{-3})

(1) 600 cm^3

(2) 400 cm^3

(3) 4000 cm^3

(4) 1400 cm^3

Answer (1)

Sol. Volume of cube inside water

$$=\left(\frac{\text{Density of cube}}{\text{Density of water}}\right)\times\text{Volume of cube}$$

$$=\frac{\text{Mass of cube}}{\text{Density of water}}$$

$$=\frac{400\text{ gm}}{1\text{ gm/cm}^3}$$

= 400 cm^3

Volume of cube outside water

= Volume of cube – Volume of cube inside water

= 1000 cm^3 – 400 cm^3

= 600 cm^3



28. A uniform rod of mass 250 g having length 100 cm is balanced on a sharp edge at 40 cm mark. A mass of 400 g is suspended at 10 cm mark. To maintain the balance of the rod, the mass to be suspended at 90 cm mark, is

(1) 290 g

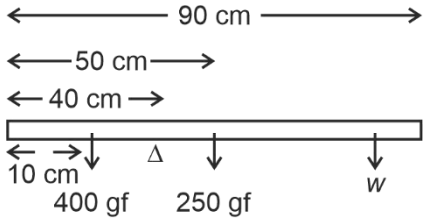
(2) 300 g

(3) 200 g

(4) 190 g

Answer (4)

Sol.



(400 gf) (30 cm) = (250 gf)(10 cm) + (wgf)(50 cm) (balancing torques)

w = 190

29. Match **List-I** with **List-II**.

	List-I		**List-II**
(A)	Angular Impulse	(I)	$[M^0 L^2 T^{-2}]$
(B)	Latent Heat	(II)	$[M L^2 T^{-3} A^{-1}]$
(C)	Electrical resistivity	(III)	$[M L^2 T^{-1}]$
(D)	Electromotive force	(IV)	$[M L^3 T^{-3} A^{-2}]$

Choose the **correct** answer from the options given below.

(1) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(2) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

(3) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(4) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

Answer (1)

Sol. [Angular Impulse] = $M L^2 T^{-1}$

[Latent Heat] = $M^0 L^2 T^{-2}$

[Electrical resistivity] = $M L^3 T^{-3} A^{-2}$

[Electromotive force] = $M L^2 T^{-3} A^{-1}$

30. In a long glass tube, mixture of two liquids *A* and *B* with refractive indices 1.3 and 1.4 respectively, forms a convex refractive meniscus towards *A*. If an object placed at 13 cm from the vertex of the meniscus in *A* forms an image with a magnification of '–2' then the radius of curvature of meniscus is

(1) 1 cm

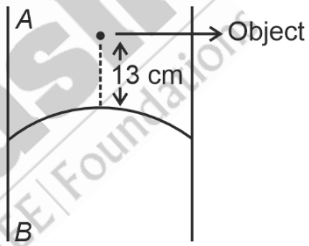
(2) $\frac{4}{3}$ cm

(3) $\frac{2}{3}$ cm

(4) $\frac{1}{3}$ cm

Answer (3)

Sol.



$\mu_A = 1.3 \quad \mu_B = 1.4$

$u = -13$ cm

$$m = \frac{\frac{v}{\mu_B}}{\frac{u}{\mu_A}} = -2$$

$$\Rightarrow \quad v = -2u\left(\frac{\mu_B}{\mu_A}\right)$$

$$= -2(-13\text{ cm})\left(\frac{1.4}{1.3}\right)$$

$$= +28\text{ cm}$$



Also,

$$\frac{\mu_B}{v}-\frac{\mu_A}{u}=\frac{\mu_B-\mu_A}{R}$$

$$\frac{1.4}{28}-\frac{1.3}{-13}=\frac{0.1}{R}$$

$$\Rightarrow \quad R=\frac{2}{3}\text{ cm}$$

31. Given below are two statements. One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Knowing initial position x_0 and initial momentum p_0 is enough to determine the position and momentum at any time t for a simple harmonic with a given angular frequency ω.

Reason (R): The amplitude and phase can be expressed in terms of x_0 and p_0.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both **(A)** and **(R)** are true but **(R)** is **NOT** the correct explanation of **(A)**

(2) **(A)** is false but **(R)** is true

(3) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(4) **(A)** is true but **(R)** is false

Answer (3)

Sol. If we express position $x(t) = A\sin(\omega t+\phi)$

then $x_0 = A\sin\phi$

$v_0 = A\omega\cos\phi$

$$\Rightarrow \quad \tan\phi=\frac{\omega x_0}{v_0}$$

$$A=\sqrt{x_0^2+\frac{v_0^2}{\omega^2}}$$

Hence both position and linear momentum of a particle can be expressed as a function of time if we know initial momentum and position

32. Earth has mass 8 times and radius 2 times that of a planet. If the escape velocity from the earth is 11.2 km/s, the escape velocity in km/s from the planet will be:

(1) 8.4

(2) 2.8

(3) 11.2

(4) 5.6

Answer (4)

Sol. $v_{escape}=\sqrt{\frac{2GM}{R}}$; $M_{earth}=8M_{planet}$; $R_{earth}=2R_{planet}$

$$\frac{v_{planet}}{v_{earth}}=\sqrt{\frac{M_{planet}}{M_{earth}}\times\frac{R_{earth}}{R_{planet}}}$$

$$v_{planet}=(11.2\text{ km/s})\sqrt{\frac{1}{8}\times 2}$$

= 5.6 km/s

33. A uniform magnetic field of 0.4 T acts perpendicular to a circular copper disc 20 cm in radius. The disc is having a uniform angular velocity of 10 π rad s^{-1} about an axis through its centre and perpendicular to the disc. What is the potential difference developed between the axis of the disc and the rim? (π = 3.14)

(1) 0.0628 V

(2) 0.5024 V

(3) 0.1256 V

(4) 0.2512 V

Answer (4)

Sol. $E=\frac{1}{2}B\omega R^2$

$$=\frac{1}{2}(0.4)(10\pi)(0.2)^2\text{ volt}$$

= 0.2512 V



34. Which of the following phenomena can not be explained by wave theory of light?

(1) Refraction of light

(2) Reflection of light

(3) Diffraction of light

(4) Compton effect

Answer (4)

Sol. Compton effect refers to scattering of a photon by free electrons. This phenomenon provides an evidence for particle nature of light.

35. The ratio of vapour densities of two gases at the same temperature is $\frac{4}{25}$, then the ratio of r.m.s. velocities will be:

(1) $\frac{25}{4}$

(2) $\frac{4}{25}$

(3) $\frac{2}{5}$

(4) $\frac{5}{2}$

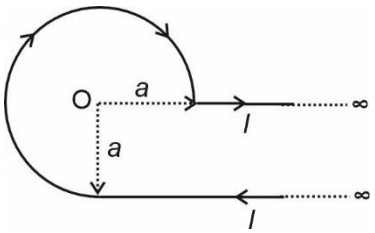
Answer (4)

Sol. (Vapor density) $=\left(\frac{\text{Molar mass}}{2}\right)$, i.e, $vd=\frac{M}{2}$

$$v_{rms}=\sqrt{\frac{3RT}{M}}$$

$$\frac{v_2}{v_1}=\sqrt{\frac{M_1}{M_2}}=\sqrt{\frac{25}{4}}=\frac{5}{2}$$

36.



An infinite wire has a circular bend or radius *a*, and carrying *a* current *I* as shown in figure. The magnitude of magnetic field at the origin O of the arc is given by:

(1) $\frac{u_0}{4\pi}\frac{I}{a}\left[\frac{3\pi}{2}+1\right]$

(2) $\frac{u_0}{2\pi}\frac{I}{a}\left[\frac{\pi}{2}+2\right]$

(3) $\frac{u_0}{4\pi}\frac{I}{a}\left[\frac{3\pi}{2}+2\right]$

(4) $\frac{u_0}{4\pi}\frac{I}{a}\left[\frac{\pi}{2}+1\right]$

Answer (1)

Sol. $B_{net}=\frac{\mu_0 I}{4\pi a}+\frac{3}{8}\frac{\mu_0 I}{a}$

$$=\frac{\mu_0 I}{4\pi a}\left(1+\frac{3\pi}{2}\right)$$

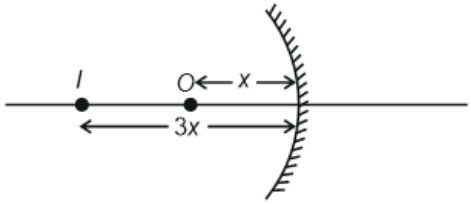
37 A concave mirror produces an image of an object such that the distance between the object and image is 20 cm. If the magnification of the image is $'-3'$, then the magnitude of the radius of curvature of the mirror is:

(1) 30 cm (2) 7.5 cm

(3) 3.75 cm (4) 15 cm

Answer (4)

Sol.



$$m=-3=-\frac{u}{v}$$

$u=-x$

$v=-3x$

$\Rightarrow$ $2x=20$ cm

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$x = 10$ cm

$$f = \frac{uv}{u+v} = \frac{(-10)(-30)}{(-10)+(-30)} = -7.5 \text{ cm}$$

$$f = -\frac{R}{2}$$

$R = -2f = 15$ cm

38. The frequency of revolution of the electron in Bohr's orbit varies with n, the principal quantum number as

(1) $\frac{1}{n^2}$

(2) $\frac{1}{n^3}$

(3) $\frac{1}{n}$

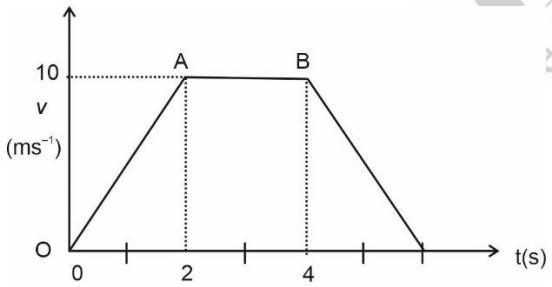
(4) $\frac{1}{n^4}$

Answer (2)

Sol. $$f = \frac{v}{2\pi r} = \left(\frac{v_0}{n}\right)\left(\frac{1}{2\pi r_0 n^2}\right)$$

$$f = \frac{v_0}{2\pi r_0} = \left(\frac{1}{n^3}\right) = f_0\left(\frac{1}{n^3}\right)$$

39. The velocity a time graph of an object moving along a straight line is shown in figure. What is the distance covered by the object between t = 0 to t = 4s?



(1) 10 m (2) 30 m

(3) 11 m (4) 13 m

Answer (2)

Sol. Distance travelled = displacement when direction of velocity remains constant

$\Rightarrow$ Distance = Area

$$= \frac{1}{2}(2s + 4s)(10 \text{ m/s})$$

$= 30$ m

40. A balloon and its content having mass M is moving up with an acceleration 'a'. The mass that must be released from the content so that the ballon starts moving up with an acceleration '3a' will be

(Take 'g' as acceleration due to gravity)

(1) $\frac{3\,Ma}{3a-g}$

(2) $\frac{3\,Ma}{2a-g}$

(3) $\frac{2\,Ma}{3a+g}$

(4) $\frac{3\,Ma}{2a+g}$

Answer (3)

Sol. Let us assure buoyancy force on balloon be B.

$B - Mg = Ma$ ...(i)

$B - (M-m)g = (M-m)3a$...(ii)

Subtracting (ii) from (i)

$(B - Mg) - (B - (M-m)g) = Ma - 3(M-m)a$

$-mg = (3m - 2M)a$

or $m = \frac{2Ma}{3a+g}$



41. The magnetic field of an E.M. wave is given by $\vec{B}=\left(\frac{\sqrt{3}}{2}\hat{i}+\frac{1}{2}\hat{j}\right)30\sin\left[\omega\left(t-\frac{z}{c}\right)\right]$ (S.I. Units). The corresponding electric field in S.I. units is

(1) $\vec{E}=\left(\frac{3}{4}\hat{i}+\frac{1}{4}\hat{j}\right)30c\cos\left[\omega\left(t-\frac{z}{c}\right)\right]$

(2) $\vec{E}=\left(\frac{1}{2}\hat{i}+\frac{\sqrt{3}}{2}\hat{j}\right)30c\sin\left[\omega\left(t+\frac{z}{c}\right)\right]$

(3) $\vec{E}=\left(\frac{1}{2}\hat{i}-\frac{\sqrt{3}}{2}\hat{j}\right)30c\sin\left[\omega\left(t-\frac{z}{c}\right)\right]$

(4) $\vec{E}=\left(\frac{\sqrt{3}}{2}\hat{i}-\frac{1}{2}\hat{j}\right)30c\sin\left[\omega\left(t+\frac{z}{c}\right)\right]$

Answer (3)

Sol. $\hat{c}=\hat{k}$

$$\hat{B}=\frac{\sqrt{3}}{2}\hat{i}+\frac{1}{2}\hat{j}$$

$$\hat{E}=\hat{B}\times\hat{C}=\frac{1}{2}\hat{i}-\frac{\sqrt{3}}{2}\hat{j}$$

$$|\vec{E}|=|\vec{B}|c=30c$$

$$\Rightarrow\quad \vec{E}=\left(\frac{1}{2}\hat{i}-\frac{\sqrt{3}}{2}\hat{j}\right)30c\sin\omega\left(t-\frac{z}{c}\right)$$

42. A parallel plate capacitor of capacitance 1 μF is charged to a potential difference of 20 V. The distance between plates is 1 μm. The energy density between plates of capacitor is.

(1) 1.8×10^{5} J/m^3

(2) 1.8×10^{3} J/m^3

(3) 2×10^{-4} J/m^3

(4) 2×10^{2} J/m^3

Answer (2)

Sol. Energy density = $\frac{1}{2}\varepsilon_0E^2$

$$=\frac{1}{2}\varepsilon_0\left(\frac{V}{d}\right)^2$$

$$=\frac{1}{2}(8.85\times10^{-12})\left(\frac{20}{10^{-6}}\right)^2\ \text{J/m}^3$$

$$\simeq 1.8\times10^{3}\ \text{J/m}^3$$

43. The kinetic energy of translation of the molecules in 50 g of CO_2 gas at 17°C is

(1) 3582.7 J

(2) 3986.3 J

(3) 4102.8 J

(4) 4205.5 J

Answer (3)

Sol. Kinetic energy of translation = $\frac{3}{2}nRT$

$$n=\frac{50\ g}{44\ g}=\frac{25}{22}\ \text{mol}$$

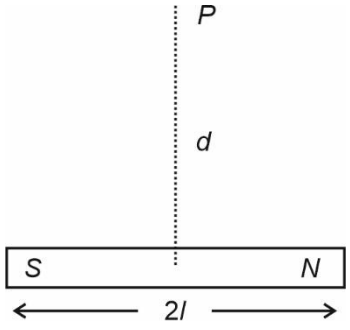
$$T=17°C=290\ K$$

$\Rightarrow$ Kinetic energy of translation

$$=\frac{3}{2}\left(\frac{25}{22}\right)(8.3)(290)\ \text{J}$$

$$=4102.8\ \text{J}$$

44.





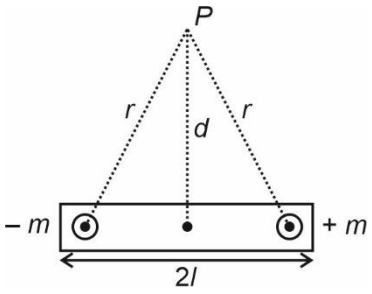
A bar magnet has total length $2l = 20$ units and the field point P is at a distance $d = 10$ units from the centre of the magnet. If the relative uncertainty of length measurement is 1%, then uncertainty of the magnetic field at point P is

(1) 10% (2) 5%

(3) 3% (4) 4%

Answer (4)

Sol.

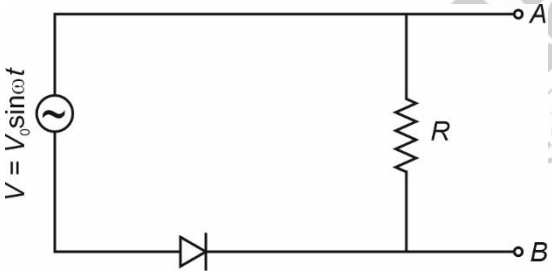


Magnetic field at P, $B = \frac{\mu_0}{4\pi}\frac{m(2l)}{r^3}$, where m is the pole strength

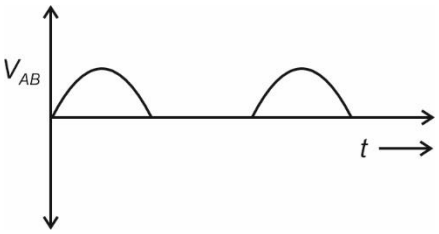
$$\Rightarrow \left(\frac{\Delta B}{B}\times 100\right) = \left(\frac{\Delta l}{l}\right)\times 100 + 3\left|\frac{\Delta r}{r}\right|\times 100$$

$$= 1\% + 3\% = 4\%$$

45. In the circuit shown here, assuming threshold voltage of diode is negligibly small, then voltage V_{AB} is correctly represented by

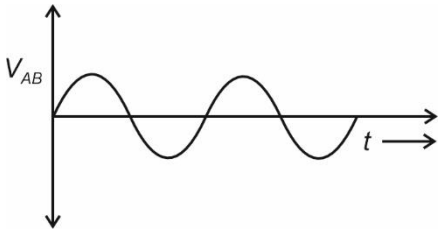


(1)

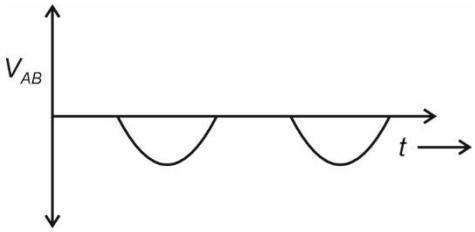


(2) V_{AB} would be zero at all times

(3)



(4)



Answer (4)

Sol. Due to the diode the current will be allowed only along B to A through the resistor. Hence V_{AB} will have only negative values *i.e.*, the following graph represents the output voltage V_{AB}.



SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

46. An electric dipole of dipole moment 6×10^{-6} Cm is placed in uniform electric field of magnitude 10^6 V/m. Initially, the dipole moment is parallel to electric field. The work that needs to be done on the dipole to make its dipole moment opposite to the field, will be _____ J.

Answer (12)

Sol. Work done in rotating a dipole = ΔU

or $W = (-PE\cos\theta_f) - (-PE\cos\theta_i)$

$= 2PE$ ($\because$ $\theta_f = 180°$ and $\theta_i = 0°$)

$= (2 \times 6 \times 10^{-6} \times 10^6)$ J = 12 J



47. The volume contraction of a solid copper cube of edge length 10 cm, when subjected to a hydraulic pressure of 7×10^6 Pa, would be _____ mm^3. (Given bulk modulus of copper = 1.4×10^{11} Nm^{-2})

Answer (50)

Sol. $B = -\frac{VP}{\Delta V}$

$\Rightarrow \Delta V = -V\frac{\Delta P}{B}$

$= -(10 \text{ cm})^3 \times \frac{7\times10^6 \text{ Pa}}{1.4\times10^{11} \text{ N/m}^2} = 50 \text{ mm}^3$

48. A thin transparent film with refractive index 1.4, is held on circular ring of radius 1.8 cm. The fluid in the film evaporates such that transmission through the film at wavelength 560 nm goes to a minimum every 12 seconds. Assuming that the film is flat on its two sides, the rate of evaporation is _____$\pi \times 10^{-13}$ m^3/s.

Answer (54)

Sol. For a thin film interference, a fringe for transmission is formed.

When

$2\mu x = n\lambda$

$\Rightarrow \frac{dx}{dt} = \left(\frac{dn}{dt}\right)\frac{\lambda}{2\mu}$

$= \left(\frac{1}{12}\right)\frac{560\times10^{-9}}{2\times1.4} = \frac{5}{3}\times10^{-8}$ m/s

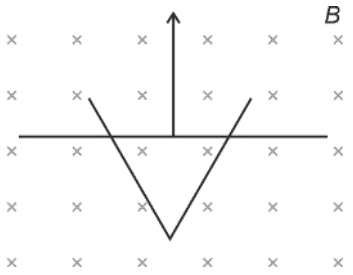
V = Volume of film = $\pi R^2 x$

$\frac{dV}{dt} = \pi R^2 \frac{dx}{dt}$

$= \pi(1.8\times10^{-2})^2 \frac{5}{3}\times10^{-8}$ m^3/s

$= 54\pi \times 10^{-13}$ m^3/s

49.



A conducting bar moves on two conducting rails as shown in the figure. A constant magnetic field B exists into the page. The bar starts to move from the vertex at time $t = 0$ with a constant velocity. If the induced EMF is $E \propto t^n$, then value of n is _____.

Answer (1)

Sol. As the bar moves without change in orientation, the length of bar will be proportional to its distance from the vertex.

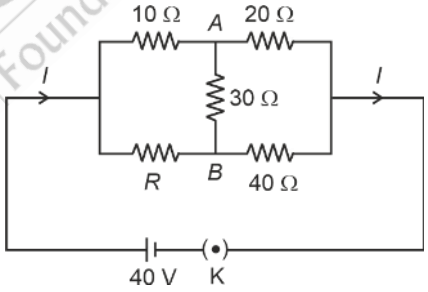
i.e. $l = c(vt)$

induced emf $E = Blv$

$= cBv^2t$

$\Rightarrow n = 1$

50. The value of current I in the electrical circuit as given below, when potential at A is equal to the potential at B, will be _____ A.



Answer (2)

Sol. Since $V_A = V_B$, the given combination is a wheatstone bridge

i.e. $\frac{R}{10\ \Omega} = \frac{40\ \Omega}{20\ \Omega} \Rightarrow R = 20\ \Omega$

$R_{eff} = \frac{(10\ \Omega + 20\ \Omega)(40\ \Omega + 20\ \Omega)}{(10\ \Omega + 20\ \Omega) + (40\ \Omega + 20\ \Omega)} = 20\ \Omega$

$\Rightarrow I = \frac{V}{R_{eff}} = \frac{40V}{20\ \Omega} = 2$ A



CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. Identify the inorganic sulphides that are yellow in colour:

(A) $(NH_4)_2S$ (B) PbS

(C) CuS (D) As_2S_3

(E) As_2S_5

Choose the correct answer from the options given below:

(1) (A), (D) and (E) only

(2) (A) and (B) only

(3) (A) and (C) only

(4) (D) and (E) only

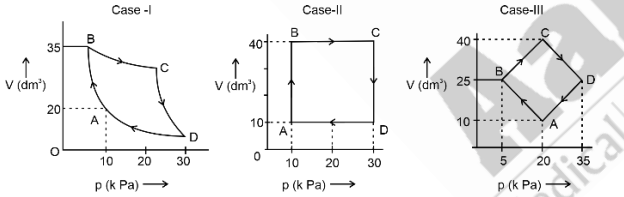
Answer (1)

Sol. $(NH_4)_2S$: colourless to yellow

PbS and CuS : Black

As_2S_3 and As_2S_5 : Yellow

52.



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path A→B→C→D→A as shown in the three cases above.

Choose the correct option regarding ΔU:

(1) ΔU(Case-I) = ΔU(Case-II) = ΔU(Case-III)

(2) ΔU(Case-III) > ΔU(Case-II) > ΔU(Case-I)

(3) ΔU(Case-I) > ΔU(Case-III) > ΔU(Case-II)

(4) ΔU(Case-I) > ΔU(Case-II) > ΔU(Case-III)

Answer (1)

Sol. As all the three case are cyclic process. Change in state functions will be zero.

Hence, $\Delta U = 0$ for all.

53. Concentrated nitric acid is labelled as 75% by mass. The volume in mL of the solution which contains 30 g of nitric acid is _____.

Given: Density of nitric acid solution is 1.25 g/mL.

(1) 55 (2) 32

(3) 45 (4) 40

Answer (2)

Sol. 75% by mass means :

$\because$ 75 g HNO_3 in 100 g solution.

$\therefore$ 30 g HNO_3 in $\frac{100}{75}\times 30$ g solution.

$$m_{solution} = \frac{100\times 30}{75}\ g$$

$$\because\ V_{sol} = \frac{m_{sol}}{d_{sol}} = \frac{100\times 30}{75\times 1.25} = 32\ mL$$

54. Identify correct conversion during acidic hydrolysis from the following:

(A) starch gives galactose.

(B) cane sugar gives equal amount of glucose and fructose.

(C) milk sugar gives glucose and galactose.

(D) amylopectin gives glucose and fructose.

(E) amylose gives only glucose.

Choose the **correct** answer from the option given below

(1) (B), (C) and (D) only (2) (A), (B) and (C) only

(3) (C), (D) and (E) only (4) (B), (C) and (E) only

Answer (4)

Sol. Starch gives glucose on hydrolysis amylose and amylopectin are components of starch.

Hence, (B), (C) and (E) are correct.

